Unsupervised Multi-hop Question Answering by Question Generation

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Introduction

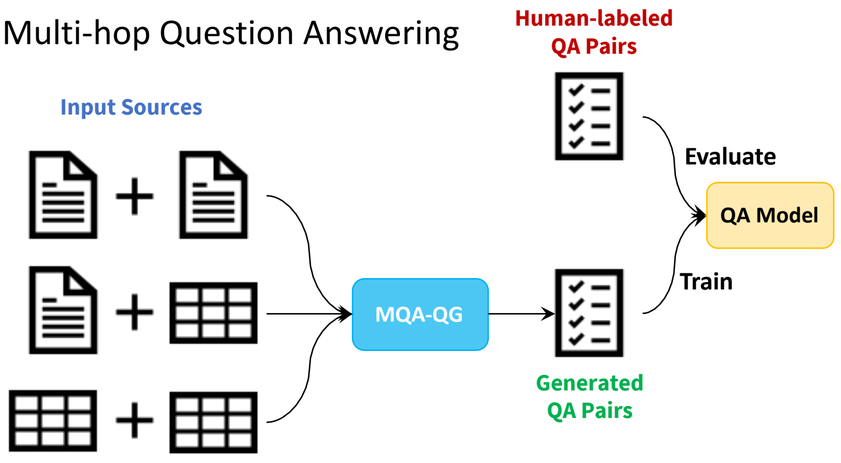
Motivation

- Training a QA system **requires large-scale data**.
- It is **time-consuming** for human to write a high-quality multi-hop question.



Could we use QG to generate synthetic QA-pairs to train the QA model?

Zero-shot Multi-hop Question Answering



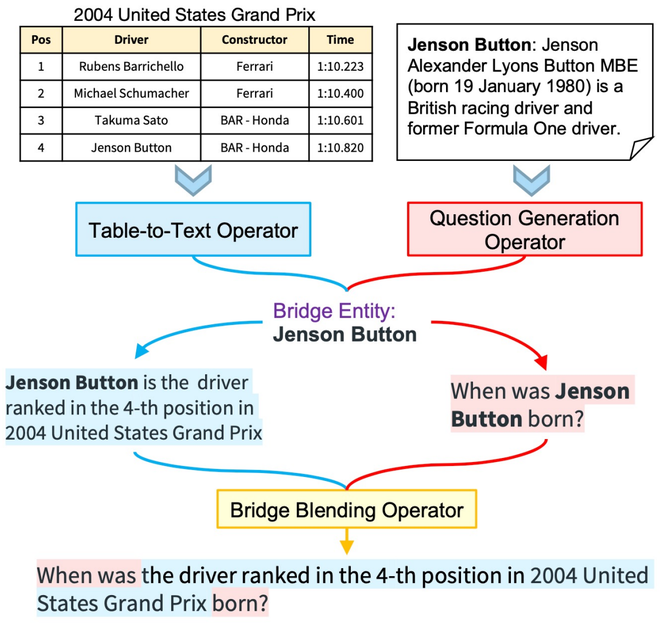
General Idea

Operators

- Retrieve / generate relevant information from a single input source
- Aggregate information from two sources.

Reasoning Graphs

- Each corresponds to one type of multi-hop question
- Formulated as a computation graph built upon the operators.

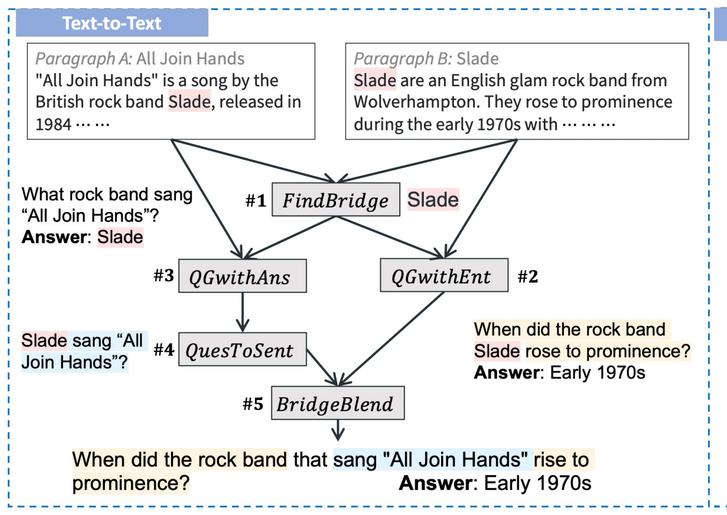


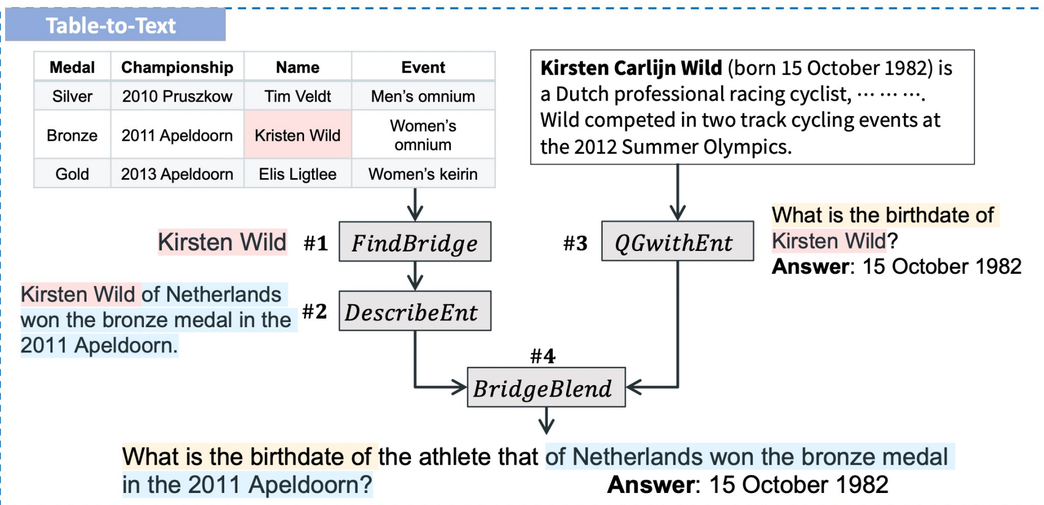
Approach

- The model first defines a set of **basic operators** to retrieve / generate relevant information from each input source or to aggregate different information, as follows.

- Afterwards, we define six **Reasoning Graphs**. Each corresponds to one type of multihop question and is formulated as a computation graph built upon the operators. We generate multihop question-answer pairs by executing the reasoning graph.

Group	Operator	Description
Selection	$FindBridge$	Select an entity $\mathcal{E}^B$ that links the two input texts $\mathcal{D}_1$ and $\mathcal{D}_2$ (or links the table $\mathcal{T}$ and the text $\mathcal{D}$)
	$FindComEnt$	Extract potential comparative entities from the input text (location, datetime, number, etc.).
Generation	$QGwithAns$	Generate a single-hop question $\mathcal{Q}$ with answer $\mathcal{A}$ from the input text $\mathcal{D}$
	$QGwithEnt$	Generate a single-hop question $\mathcal{Q}$ that contains the given entity $\mathcal{E}$ from the input text $\mathcal{D}$
	$DescribeEnt$	Generate a sentence $\mathcal{S}$ that describes the given entity $\mathcal{E}$ based on the information of the table $\mathcal{T}$
	$QuesToSent$	Convert a question $\mathcal{Q}$ into its declarative form $\mathcal{S}$
Fusion	$BridgeBlend$	Generate a bridge-type multi-hop question $\mathcal{Q}^B$ by fusing the single-hop question Q and the sentence S given the entity $\mathcal{E}^B$ as the bridge
	$CompBlend$	Generate a comparison-type multi-hop question $\mathcal{Q}^C$ by fusing two single-hop questions





Experiments

Datasets

❑ HotpotQA

Text + Text

Paragraph A: Walchelin de Ferriers

Walchelin de Ferrieres (or Walkelin de Ferrers) (died 1201) was a Norman baron and principal captain of King Richard I of England.

Paragraph B: Richard I of England

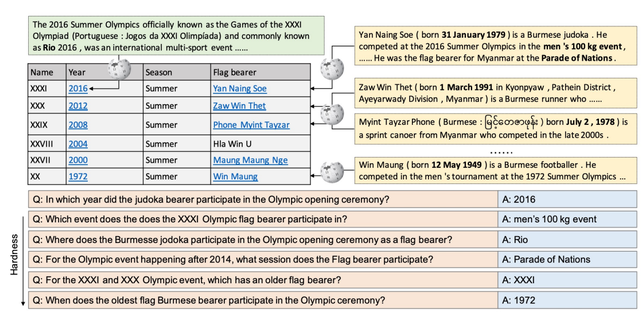
Richard I (8 September 1157 – 6 April 1199) was King of England from 6 July 1189 until his death. He also ruled as Duke of Normandy, Aquitaine and Gascony, Lord of Cyprus, Count of Poitiers, Anjou, Maine, and Nantes, and was overlord of Brittany at various times during the same period. He was the third of five sons of King Henry II of England and Duchess Eleanor of Aquitaine. He was known as Richard Cœur de Lion or Richard the Lionheart because of his reputation as a great military leader and warrior. He was also known in Occitan as Oc e No ("Yes and No"), because of his reputation for terseness.

Q: Who is the mother of the king that Walchelin de Ferriers was principal captain to?

A: Duchess Eleanor of Aquitaine

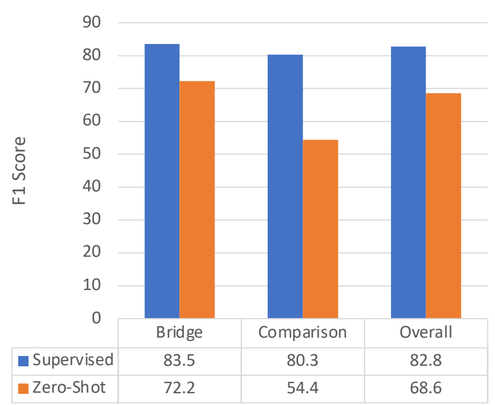
❑ HybridQA

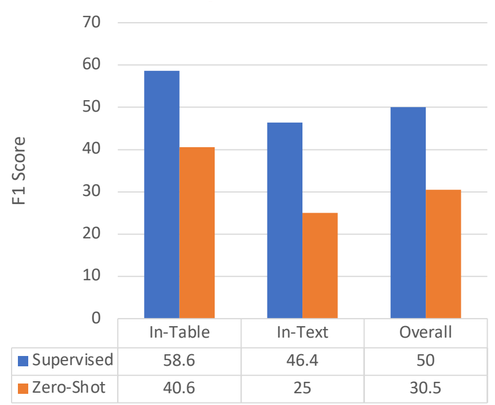
Table + Text



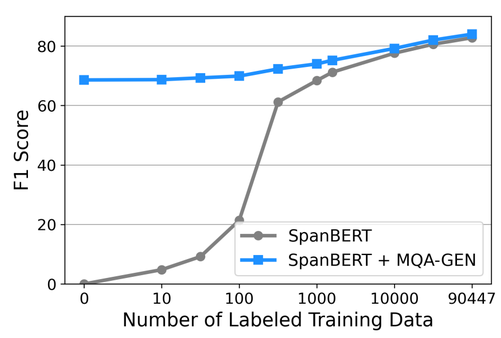
Main Results

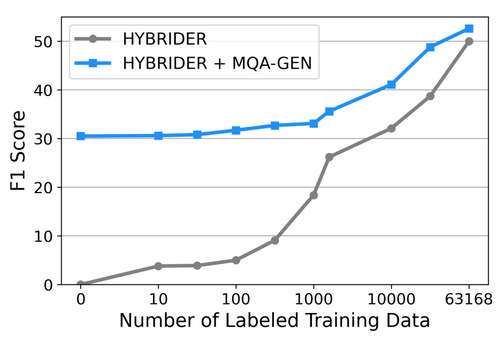
Zero-shot Multi-hop Question Answering





Few-shot Multi-hop Question Answering





Contributions

- This is the first work to investigate unsupervised multi-hop QA.
- We propose MQA-QG, a novel framework to generate high-quality training data without the need to see any human-annotated multi-hop question.
- We show that the generated training data can greatly benefit the multi-hop QA system in both unsupervised and few-shot learning settings.

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